

Lecture Assignment 2:

Qualitative and Quantitative Research Methods:

Credit Card Fraud Detection

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1 Introduction

For this lecture assignment, we are tasked with discussing how our data science research task can be solved by using qualitative or quantitative research methods. Hence, we will discuss and reflect on the possible outcomes and conclusions of each method.

2 Qualitative Research Analysis

Qualitative research can be used to understand how credit card fraud is identified and experienced in real situations. Instead of using numerical transaction data, information can be collected through interviews with people who work with fraud detection. The research question could be "How do fraud detection professionals identify and understand suspicious credit card transactions?" The data needed for this method could be collected from various sources, such as interviews, articles, reports, and online sources. Information from open database websites such as Kaggle could also be used to get a better understanding of credit card fraud and the dataset. Interviews with customers or people working in fraud detection could provide additional insights into their experiences and common problems with credit card fraud. The possible outcome would be a better understanding of how fraud is detected in practice. The interviews could also show common patterns or problems that may not be visible from transaction data alone.

3 Quantitative Research Analysis

Quantitative research can be used to study credit card fraud using numerical data. For this method, we can use the Credit Card Fraud Detection dataset from Kaggle [1]. It contains a total of 284,807 transactions, 492 of which are fraudulent. The data includes information on the amount, time, class, and 28 different anonymized (V) attributes. The research question could be "Can we use transaction data to identify credit card fraud?" The data would then be used to compare normal and fraudulent transactions and find possible patterns. It's possible to use a machine learning model to predict whether a transaction is fraudulent. The system could be used by banks to find possible fraud and protect their customers. If a suspicious transaction is detected, the bank could review it and contact the customer. The possible outcome is to understand which patterns are associated with fraud and how well the model can detect fraudulent transactions. Statistical testing can be used to determine whether there is a difference between normal and fraudulent transactions.

4 Conclusions

Qualitative research analysis shows how fraud analysts interpret suspicious transactions and whether to verify, review, or block a payment. Quantitative research analysis measures how well transaction data distinguishes fraudulent payments from valid ones, as well as how often a model raises false alarms. Together, both types of methods help a card issuer design a system that supports both fraud detection

and the decisions that follow an alert. However, our selected dataset can only support quantitative analysis. Studying analysts' decisions would require interviews or investigation records.

References

- [1] kaggle.com. *Credit Card Fraud Detection*. Last accessed 27 september 2026. 2021.